

## ZERODHA CLONE INTERVIEW GUIDE

Ye PDF Zerodha Clone ke liye Hinglish interview notes hai. Isme MERN architecture, JWT auth, dashboard flow aur important code explanation diya gaya hai.

### 1. One-line Introduction

Zerodha Clone ek MERN-based full-stack stock broker style application hai jisme public landing page, login/signup aur protected portfolio dashboard implement kiya gaya hai.

### 2. Short Interview Intro

Ye project maine portfolio-strength full-stack application ke roop me build kiya. Isme React landing page, separate dashboard app, Express backend, MongoDB data storage aur JWT cookie-based authentication use ki gayi hai. User holdings, positions aur orders related dashboard features dekh sakta hai.

### 3. Tech Stack

- React
- React Router
- Axios
- Chart.js
- Node.js
- Express
- MongoDB
- Mongoose
- JWT
- bcryptjs

### 4. Main Features

- Signup and login
- Password hashing
- Cookie-based JWT auth
- Protected dashboard route
- Holdings and positions data
- Orders create flow
- Watchlist and buy modal
- Dashboard username

### 5. Architecture

Project 3 parts me split hai:

- frontend public pages
- dashboard private app UI
- backend REST APIs

### 6. Important Code with Line Meaning

Snippet A: CORS and credentials

```
const allowedOrigins = [ // Allowed frontend/dashboard domains ki list
```

```

"http://localhost:3000", // Local landing page
"http://localhost:3001", // Local dashboard
"https://zerodha-clone-landingpage.onrender.com", // Hosted frontend
"https://zerodha-dashboard.onrender.com", // Hosted dashboard
];

app.use(
  cors({
    origin: allowedOrigins, // Sirf trusted origins allowed hain
    credentials: true, // Cookie-based auth requests allow hong
  })
);

```

#### Snippet B: signup

```

const normalizedEmail = email?.trim().toLowerCase(); // Email normalization
const normalizedUsername = username?.trim(); // Username cleanup

const existingUser = await UserModel.findOne({ email: normalizedEmail }); // Duplicate check
if (existingUser) {
  return res.status(409).json({ message: "User already exists", success: false }); //
  Conflict response
}

const user = await UserModel.create({ // User creation
  email: normalizedEmail,
  password,
  username: normalizedUsername,
  createdAt,
});

```

#### Snippet C: password hashing

```

UserSchema.pre("save", async function () { // Save se pehle hook chalega
  if (!this.isModified("password")) { // Agar password change nahi hua
    return; // To rehash nahi hoga
  }

  this.password = await bcrypt.hash(this.password, 12); // Secure hash create hua
});

```

#### Snippet D: session verification route use in dashboard

```

useEffect(() => {
  API.post("/") // Backend verify endpoint hit hota hai
  .then((res) => {
    if (res.data.status) { // Session valid
      localStorage.setItem("username", res.data.user); // Username save hua
      setStatus("auth"); // Dashboard allow hoga
    } else {
      setStatus("unauth"); // Redirect condition
    }
  })
});

```

```

    .catch(() => {
      setStatus("unauth"); // Error ko unauthorized treat kiya
    });
  }, []);

```

Snippet E: axios instance

```

const API = axios.create({
  baseURL: process.env.REACT_APP_API_BASE_URL || "http://localhost:3002", // Backend URL
  central place par
  withCredentials: true, // Cookie requests allow
});

```

Snippet F: order creation

```

API.post("/newOrder", {
  name: uid, // Stock symbol/name
  qty: stockQuantity, // Quantity input
  price: stockPrice, // Price input
  mode: "BUY", // Order mode
})
  .then(() => {
    generalContext.closeBuyWindow(); // Success par modal close
  })
  .catch(console.error); // Failure log

```

## 7. Interview Questions

Q. JWT cookie auth kyun?

Answer:

Cross-app login flow aur stateless verification ke liye ye good fit tha. Browser cookie automatically request ke sath send hoti hai.

Q. Separate dashboard kyun?

Answer:

Public marketing site aur private product dashboard ko separate rakhna real-world product structure ko better represent karta hai.

Q. Context kyun use kiya?

Answer:

Buy modal state ko watchlist se trigger karne ke liye context ne prop drilling avoid ki.

## 8. HR Style Answer

Ye project maine isliye banaya kyunki mujhe ek strong portfolio project chahiye tha jisme frontend, backend, auth aur dashboard sab practical form me ho.

## 9. Improvements

- Protected APIs par auth middleware
- Better secure cookies
- Better validation
- Dynamic watchlist

- Testing